Bayesian Framework for Canonical factorization of multiblock data

1 Introduction

Data sets structured into blocks of variables measured on the same set of observations are frequently encountered in practice. These blocks may consist of quantitative variables originating from distinct measurement sources or correspond to thematic groupings defined by the user. For instance, in omics studies, blocks typically comprise variables derived from different analytical techniques (e.g., proteomics, metabolomics). In psychology, multiblock data structures commonly arise when multiple sets of variables are collected on the same individuals to capture different psychological domains. For example, distinct blocks may include self-report questionnaire scores assessing personality traits, behavioral task performance measures, and neurophysiological indicators such as electroencephalography features.

The four expected objectives of exploratory multiblock data analysis are: (i) to perform a joint dimensionality reduction, (ii) to examine relationships among blocks, (iii) to investigate associations among variables both within and across blocks, and (iv) to capture within-block variability in order to reveal shared and block-specific structures in a common latent space. An unified algebraic framework based on factorization strategies, denoted *canonical factorisation*, was proposed by [Bougeard et al., 2024] for multiblock methods to achieve these objectives. In practice, canonical factorisation provides a structured framework for analysing data variability at multiple levels: differences between and within blocks, differences between observations by projecting them onto a common subspace. It also facilitates the harmonisation of outputs produced by different methods and supports a more objective criterion-based selection among competing approaches.

This framework is a compass among multiblock models: Generalized Canonical Correlation Analysis (GCCA, [Carroll, 1968]) and its popular case MAXVAR [Carroll, 1968], Hierarchical Principal Component Analysis (HPCA, [Wold et al., 1996, Westerhuis et al., 1998]), Structuration de Tableaux A Tois Indices de la Statistique (Statis, [Lavit et al., 1994]), Multiple Factorial Analysis (MFA, [Escofier and Pages, 1994]), Common Components and Specific Weights Analysis (CCWSA, [Qannari et al., 2001]), Multiple CO-inertia Analysis (MCOA, [Chessel and Hanafi, 1996]) fall within this framework. A benchmark of these methods had been done in [Bougeard et al., 2024].

Initially, MCOA was proposed as an alternative to generalized canonical analysis, also known as MAXVAR [Carroll, 1968, Horst, 1961], for correlating multiple data tables. Compared with other generalisations of Hotelling's canonical correlation analysis [Hotelling, 1935], such as those proposed by [Kettenring, 1971], MCOA offers three main advantages. Firstly, it is computationally straightforward as it reduces to a singular value decomposition, the other approaches typically rely on iterative optimisation procedures. Secondly, MCOA is particularly effective at summarizing results across multiple datasets. Finally, MCOA provides sequential estimation of canonical factorisation under an orthogonality constraint. In this work, we propose a new method, Optimal Multiple CO-inertia Analysis (OMCOA), which is a simultaneous version of MCOA. We compare this model with the previous one using simulated data whose structure is known.

2 Canonical factorisation

Notations

- K blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ respectively of $p_1, \ldots, p_K$ variables on N individuals.
- $\mathbf{X} = [\mathbf{X}_1 | \ldots | \mathbf{X}_K] \in \mathcal{M}_{N,P}(\mathbb{R})$ $(P = \sum_{k=1}^{K} p_k)$ be the concatenation of these K-blocks.
- $\mathbf{U}^\intercal$ designates the transposed matrix of $\mathbf{U}$.
- $\Sigma_k = diag(\sigma_k^{(1)}, ..., \sigma_k^{(R)})$ are diagonal matrices
- $\|.\|$ correspond to the Frobenius norm.

Consider N individuals characterized by K blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ respectively of $p_1, \ldots, p_K$ variables. Let $\mathbf{X} = [\mathbf{X}_1 | \ldots | \mathbf{X}_K] \in \mathcal{M}_{N,P}(\mathbb{R})$ $(P = \sum_{k=1}^{K} p_k)$ be the concatenation of these K-blocks. Without loss of generality, the blocks are assumed to be column-centered.

The $(K+1)-$blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K, \mathbf{X})$ can be factorized in dimension $R = rank(\mathbf{X})$ as follows (see [Bougeard et al., 2024] for the demonstration)

$$\left\{ \begin{array}{l} \mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^\intercal \\ \mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal \quad \text{for} \quad k = (1, \ldots, K) \\ \text{with the constraints} \quad \mathbf{U}^\intercal\mathbf{U} = \mathbf{I}_R \quad \text{and} \quad \|\mathbf{v}_k^{(1)}\| = \ldots = \|\mathbf{v}_k^{(R)}\| = 1 \quad \text{for} \quad k = (1, \ldots, K), \end{array} \right. \tag{1}$$

where

- $\mathbf{U} \in \mathcal{M}_{N,H}(\mathbb{R})$ is the standardized and orthogonal common components,
- $\mathbf{V} \in \mathcal{M}_{P,H}(\mathbb{R})$ is the standardized (not necessarily orthogonal) common loadings,
- $\mathbf{V}_k \in \mathcal{M}_{p_k,H}(\mathbb{R})$ the standardized block-loadings, $\Sigma_1 \in \mathcal{M}_{R,R}(\mathbb{R})$
- $\Sigma \in \mathcal{M}_{R,R}(\mathbb{R})$ the diagonal scaling matrices with property $\Sigma^2 = \sum_{k=1}^{K} \Sigma_k^2$ that relate common and block-analyses.

This factorization is *the canonical factorization*. Each multiblock method derives a common component $\mathbf{U}$ using a different optimization criterion. Projecting the data onto these common components enables their canonical factorization (for details see [Bougeard et al., 2024]). The benchmark in [Bougeard et al., 2024] showed that all the methods under study produced similar results. The decomposition of the concatenated block $\mathbf{X}$ is unique for all the multiblock methods, however, only the MCOA method guarantees uniqueness of the decomposition of each block $\mathbf{X}_k$ (thanks to the orthoganlity constraint), and the proportion of variance it explains can be lower than that of the other methods. To ensure uniqueness of the solution and to improve the robustness of MCOA, we propose a simultaneous version.

For $H < rank(\mathbf{X})$, the objective is to determine the optimal approximation of the form given in (1) by minimizing an ordinary least square criterion. The models in section 1 (GCCA, SUMCOR, HPCA, Statis, CCWSA, MCOA) provide solution of this problem (see [Bougeard et al., 2024] for details on these methods).

3 Bayesian framework for Canonical Factorisation

In the following, we assume that for all $k \in 1, \ldots, K$ the columns of $\mathbf{V}_k$ are orthogonal i.e.

$$\mathbf{V}_k^\intercal \mathbf{V}_k = \mathbf{I}_R.$$

3.1 Minimization problem

Providing a solution for canonical factorization means minimizing

$$\sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal\|^2 = \|\mathbf{X} - \mathbf{U}\Sigma\mathbf{V}^\intercal\|^2, \tag{2}$$

with $\mathbf{U}^\intercal.\mathbf{U} = I_R$, $\mathbf{V}_k^\intercal.\mathbf{V}_k = I_R$ for $k = 1, \ldots, K$. Note that, with standardized and orthogonal block-loadings matrices $\mathbf{V}_1, \ldots, \mathbf{V}_K$, the solution of this problem is unique.

Consider $\mathbf{H}_k = \Sigma_k.\mathbf{V}_k^\intercal \in \mathcal{M}_{R,p_k}(\mathbb{R})$ and $\mathbf{H} = [\mathbf{H}_1|\ldots|\mathbf{H}_K] \in \mathcal{M}_{R,P}(\mathbb{R})$. An iterative algorithm for building the solution of (2) requires to update successively $\mathbf{U}, \mathbf{V}_k, \Sigma_k$ as follows:

1. Updating of $\mathbf{U}$: For given $\mathbf{X} \in \mathcal{M}_{N,P}(\mathbb{R})$, $\mathbf{H} \in \mathcal{M}_{R,P}(\mathbb{R})$, update $\mathbf{U}$ by $\mathbf{U} = \mathbf{A}.\mathbf{B}^\intercal$ where $\mathbf{A}, \mathbf{B}$ are respectively the matrices of the first R left singular vectors and the matrix of the first R right singular vectors from the svd of $\mathbf{X}.\mathbf{H}^\intercal$.

2. Updating of $\mathbf{V}_k$: For given $\mathbf{X}_k \in \mathcal{M}_{N,p_k}(\mathbb{R})$, $\mathbf{U} \in \mathcal{M}_{R,N}(\mathbb{R})$ with orthogonal columns, $\Sigma_k \in \mathcal{M}_{R,R}(\mathbb{R})$, update $\mathbf{V}_k$ by: $\mathbf{V}_k = \mathbf{A}\mathbf{B}^\intercal$ where $\mathbf{A}, \mathbf{B}$ are respectively the matrix of the first R left singular vectors and the matrix of the first R right singular vectors from the svd of $\mathbf{X}_k^\intercal.\mathbf{U}$.

3. Updating of Σ_k: For given $\mathbf{X}_k \in \mathcal{M}_{N,p_k}(\mathbb{R})$, $\mathbf{U} \in \mathcal{M}_{R,N}(\mathbb{R})$ orthogonal, $\mathbf{V}_k \in \mathcal{M}_{R,p_k}(\mathbb{R})$ with orthonormal columns, Σ_k is given by

$$\Sigma_k = diag(\mathbf{U}^\intercal \mathbf{X}_k \mathbf{V}_k).$$

Proposition 1. *The successive updates of $\boldsymbol{U}$, $\boldsymbol{V}_k$, and Σ_k guarantee that the criterion defined in* (2) *decreases or remains constant at each step, ensuring convergence of the algorithm.*

4 Comparison of MCOA and OMCOA

4.1 Simulation study

MCOA and OMCOA are compared on simulated data. $K = 3$ blocks $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3$ with respectively $p_1 = 10$, $p_2 = 15$, $p_3 = 20$ variables and $N = 100$ observations are generated from

$$\mathbf{X}_k = \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal + \mathbf{E}_k, \qquad k = 1, 2, 3,$$

where $\mathbf{U} \in \mathcal{M}_{N,R}(\mathbb{R})$ (column-centered) and $\mathbf{V}_k \in \mathcal{M}_{p_k,R}(\mathbb{R})$ have orthonormal columns, $R = 3$, Σ_k is diagonal and $\mathbf{E}_k$ is a Gaussian noise term. We compare the estimation of $\Sigma_1, \ldots, \Sigma_K$ obtained with MCOA and OMCOA. The diagonal matrices depend on a parameter $t \in [0, 1]$:

$$\Sigma_k(t) = (1 - t)\, \Sigma_k^{(0)} + t\, \Sigma_k^{(1)},$$

where

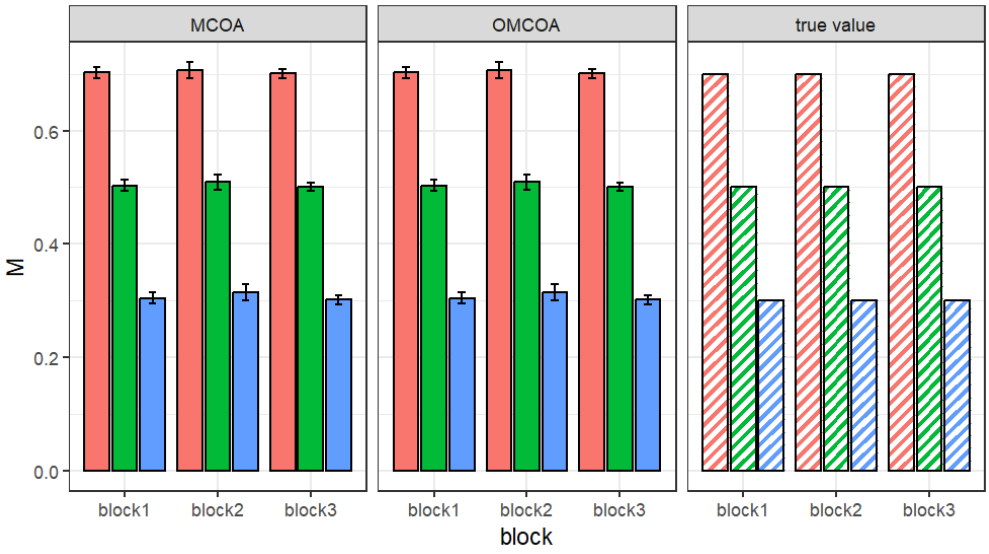


Figure 1: Common profile ($t = 0$): median and 10–90% empirical quantiles of $\hat{\sigma}_k^{(\hat{\pi}(r))}$ over the $B = 1000$ replications (red: dim 1, green: dim 2, blue: dim 3).

- for $t = 0$, all the blocks share the same profile: $\Sigma_1^{(0)} = \Sigma_2^{(0)} = \Sigma_3^{(0)} = \text{diag}(0.7, 0.5, 0.3)$;
- for $t = 1$, each block has its own dominant dimension of the common space $\mathbf{U}$ (permuted profiles): $\Sigma_1^{(1)} = \text{diag}(0.7, 0.5, 0.3)$, $\Sigma_2^{(1)} = \text{diag}(0.5, 0.3, 0.7)$, $\Sigma_3^{(1)} = \text{diag}(0.3, 0.7, 0.5)$.

In both cases $\sum_r (\sigma_k^{(r)})^2 = 0.83$ for each block. Moreover, the singular values of the noise-free concatenated matrix are $\left(\sum_k (\sigma_k^{(r)}(t))^2\right)^{1/2}$: they are well separated for $t = 0$ and all equal to $\sqrt{0.83}$ for $t = 1$, since $\sum_k {\Sigma_k^{(1)}}^2 = 0.83\,\mathbf{I}_3$. The noise is scaled so that $\|\mathbf{E}_k\|^2 = \frac{0.10}{0.90}\,\|\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$, i.e. it accounts for about 10% of the variance of each block. We use $t \in \{0, 0.1, \ldots, 1\}$ and $B = 1000$ replications.

At each replication, the estimated axes are reordered by the permutation

$$\hat{\pi} = \arg\max_{\pi \in \mathfrak{S}_R} \sum_{r=1}^{R} \left|\cos\left(\mathbf{u}^{(r)}, \hat{\mathbf{u}}^{(\pi(r))}\right)\right|,$$

the absolute value accounting for the sign indeterminacy of the axes. The accuracy of the estimation of $\Sigma_1, \ldots, \Sigma_K$ is assessed by the mean absolute error over the K blocks and the R dimensions:

$$\text{MAE} = \frac{1}{KR} \sum_{k=1}^{K} \sum_{r=1}^{R} \left|\hat{\sigma}_k^{(\hat{\pi}(r))} - \sigma_k^{(r)}\right|.$$

4.2 Results

Figures 1 and 2 show the estimates obtained with MCOA and OMCOA for $t = 0$ and $t = 1$ respectively. As expected, both methods give very close estimates when the blocks share a common profile (Figure 1). When the profiles are permuted, OMCOA outperforms MCOA in terms of both bias and precision (Figure 2). In both scenarios, the OMCOA estimates show very low variability.

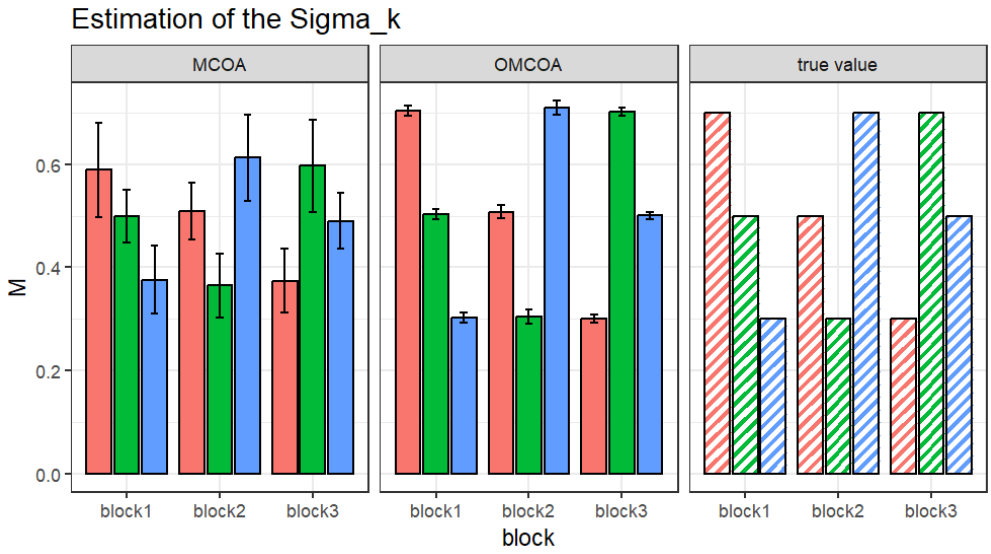


Figure 2: Permuted profiles ($t = 1$): median and 10–90% empirical quantiles of $\hat{\sigma}_k^{(\hat{\pi}(r))}$ over the $B = 1000$ replications (red: dim 1, green: dim 2, blue: dim 3).

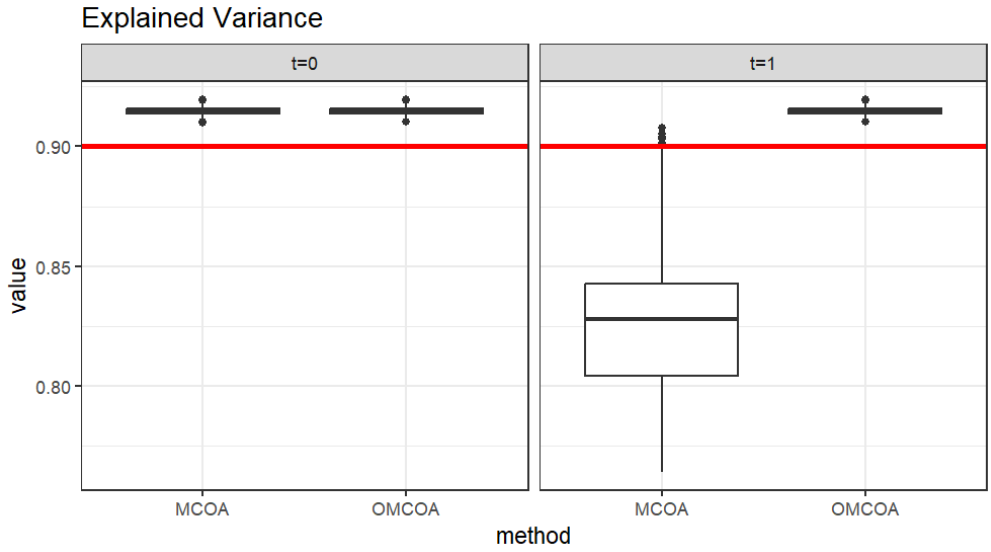


Figure 3: Distribution of the percentage of explained variance V over the $B = 1000$ replications.

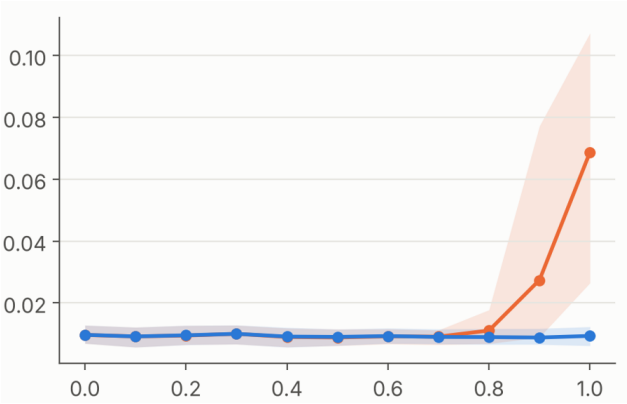


Figure 4: Median and 10–90% empirical quantiles of the MAE as a function of t ($B = 1000$ replications).

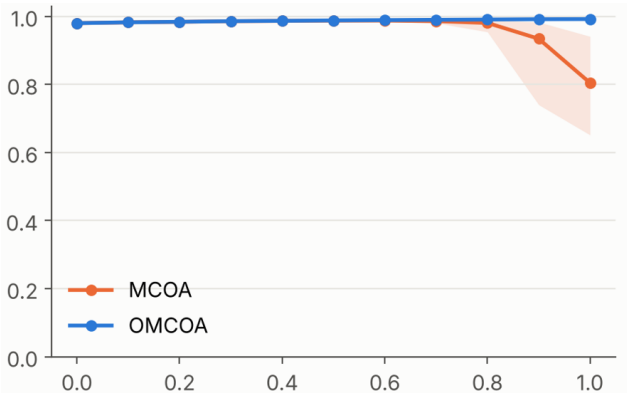


Figure 5: Median and 10–90% empirical quantiles of the worst axis recovery $\min_r |\cos(\mathbf{u}^{(r)}, \hat{\mathbf{u}}^{(\hat{\pi}(r))})|$ as a function of t ($B = 1000$ replications).

Figure 3 shows the empirical distribution of the percentage of variance of the concatenated matrix $\mathbf{X}$ explained by the R dimensions,

$$V = 100 \times \frac{\sum_{k,r} (\hat{\sigma}_k^{(r)})^2}{\|\mathbf{X}\|^2}, \qquad \|\mathbf{X}\|^2 = \sum_{k=1}^{K} \|\mathbf{X}_k\|^2,$$

to be compared with the percentage of variance due to the signal, which is about 90% by construction. OMCOA slightly exceeds this value in all cases, since part of the noise is captured by the fitted dimensions, whereas MCOA clearly underestimates it when the profiles are permuted ($t = 1$).

Figure 4 shows the distribution of the MAE as a function of t. MCOA and OMCOA coincide as long as the singular values of $\mathbf{X}$ are well separated; MCOA breaks down when they get close ($t \geq 0.8$). Indeed, MCOA starts with the SVD of $\mathbf{X}$: when the singular values are close, $\mathbf{u}^{(1)}$ is an arbitrary direction of span($\mathbf{U}$) driven by the noise, and this error propagates to the next dimensions through the deflation. OMCOA fits all the blocks simultaneously, so that the block-specific profiles identify $\mathbf{U}$. Figure 5 illustrates the resulting misestimation of the axes $\mathbf{u}^{(1)}, \ldots, \mathbf{u}^{(R)}$ by MCOA for $t \geq 0.8$.

Conclusion

[Bougeard et al., 2024] have compared estimations for the canonical factorisation problem. In this work, we propose on optimal version of Multiple COinertia Analysis that performs better on the simulated data. Constraining the matrices $(\mathbf{V}_1, \ldots, \mathbf{V}_K)$ to be orthogonal ensures uniqueness of the solution. In particular, if $|\mathbf{E}_k| = 0$, the OMCOA method recovers the exact canonical factorization while other methods do not guarantee this." Furthermore, in practical applications the canonical factorisation is more readily interpreted under orthogonal configurations, as these yield a clearer decomposition of the effects.

References

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